

Applied Probabilistic Models

Exercise 3

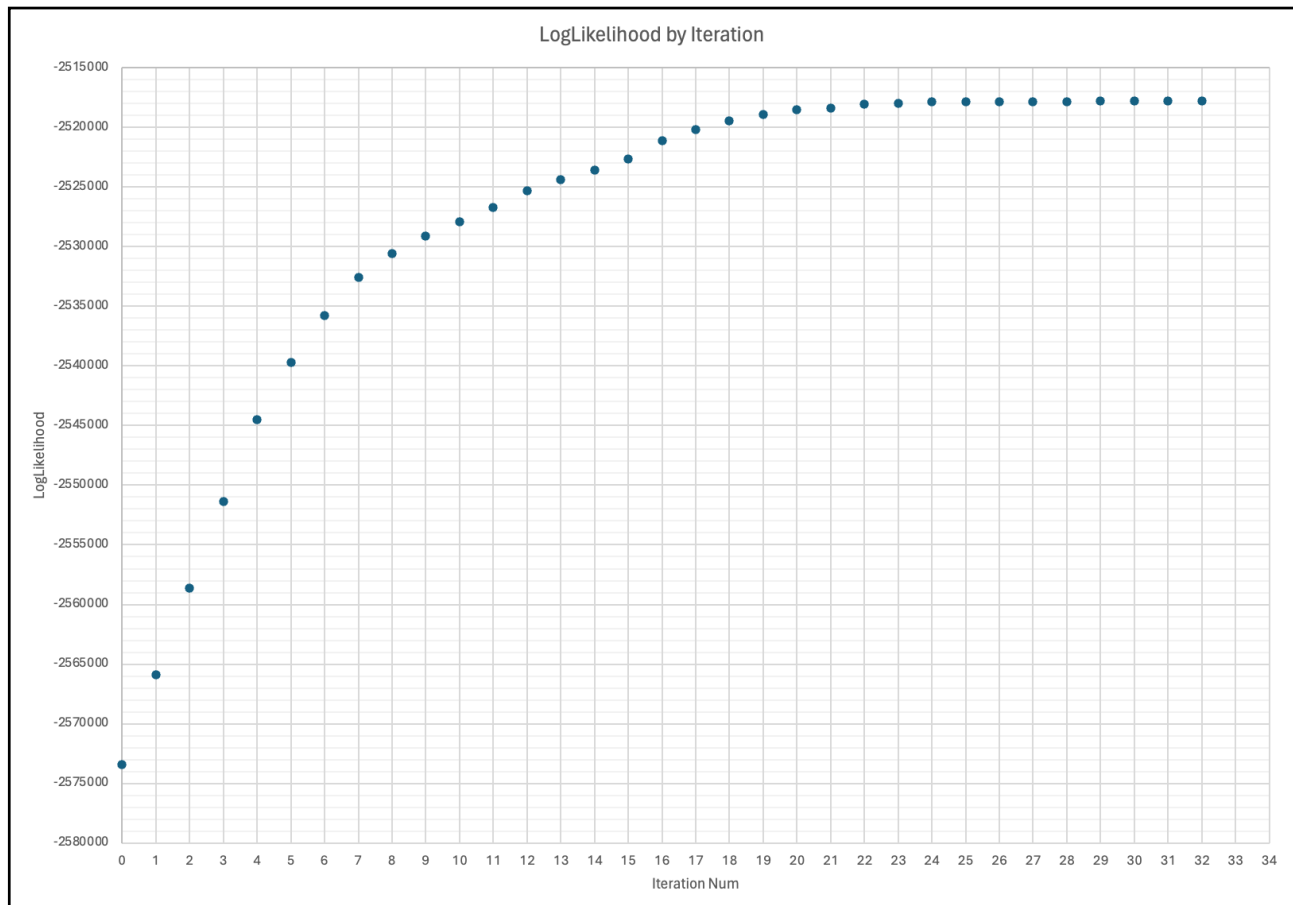
Author: Kobie Hazon

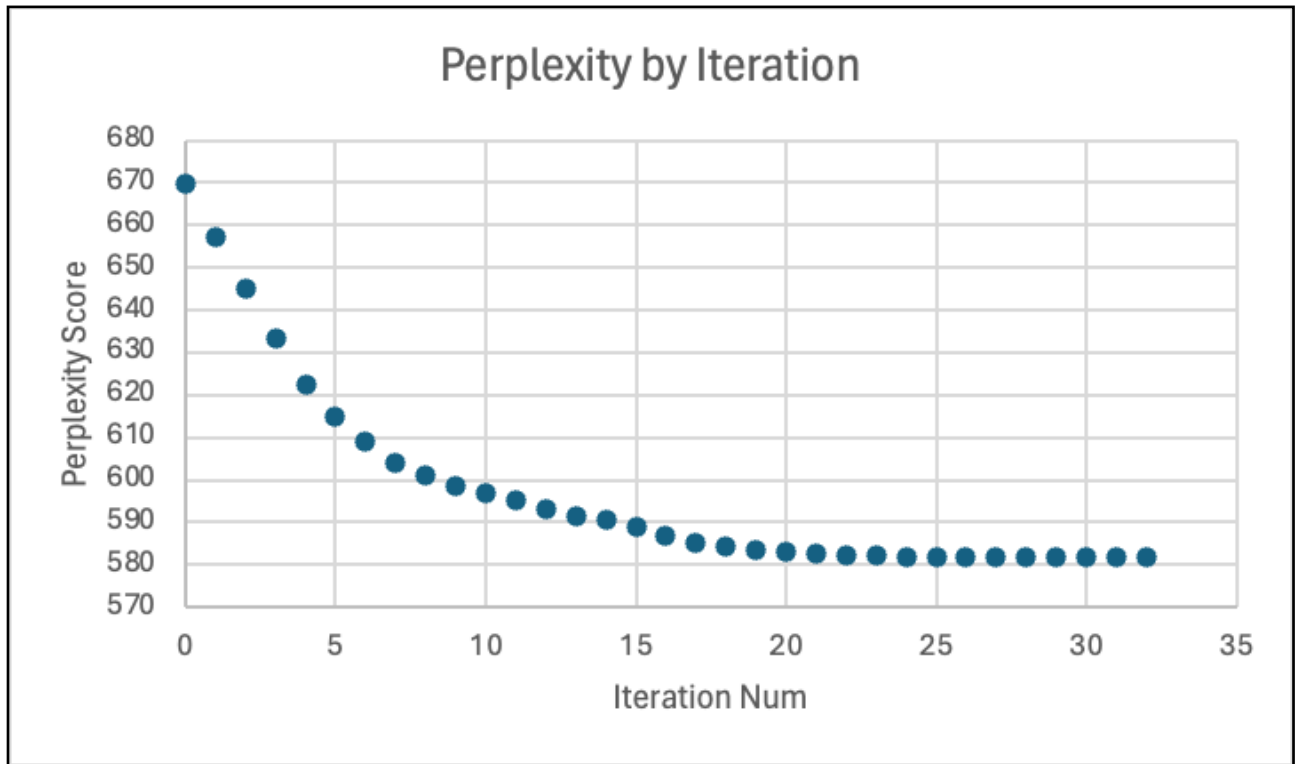
Question 1:

The threshold I chose for my implementation of the EM clustering algorithm has 3 conditions:

- Total number of iterations must be above a certain constant (I chose 5) - this condition's purpose is to make sure that the algorithm had the chance to make advancements even if the cluster's initialization is on a "stale" spot.
- Distance between the In-likelihood of this iteration and the last one is less then the STOP_LN_LIKELIHOOD_THRESHOLD constant (I chose 5 for this too). In terms of the In-likelihood, 5 is a very small number - this condition makes sure that were are very close to a local maxima s.t. our progression is very slow and not computationally efficient.
- Distance between the In-likelihood of the last iteration and the prior one is less then STOP_LN_LIKELIHOOD_THRESHOLD - this is done to be more certain that we are near a local maxima and the last condition wasn't just a singular slow down in our advancement.

Question 2:



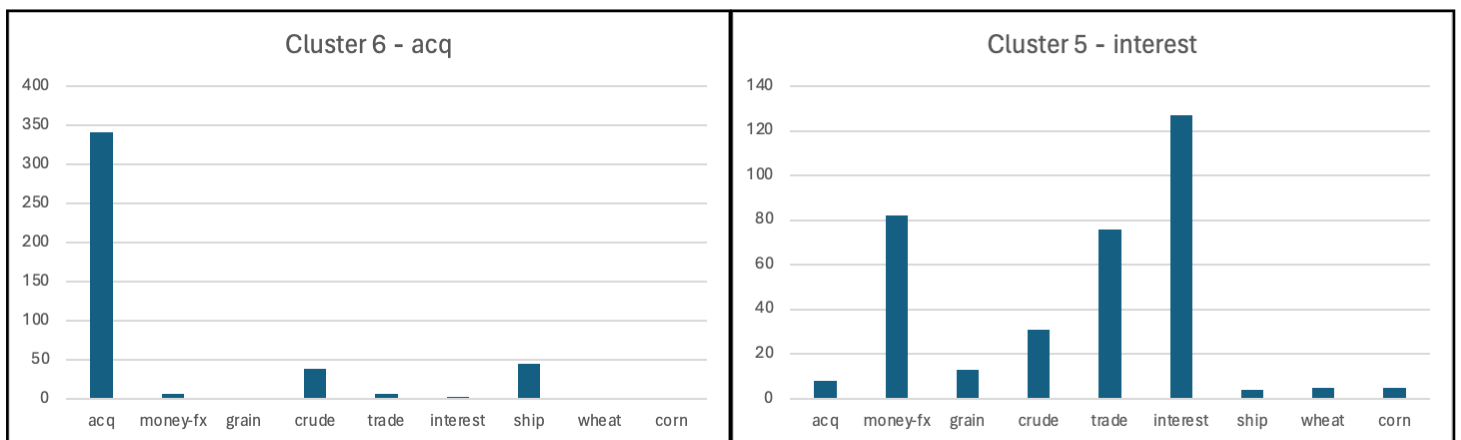


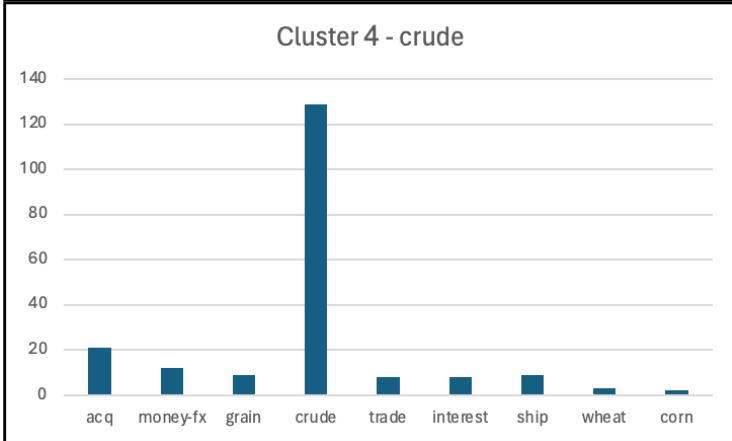
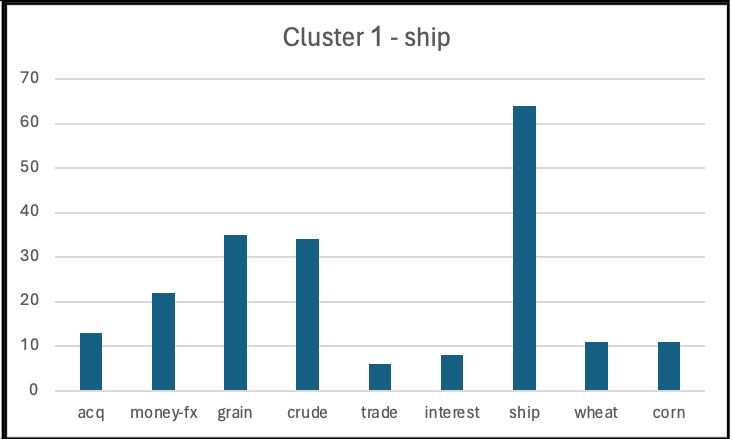
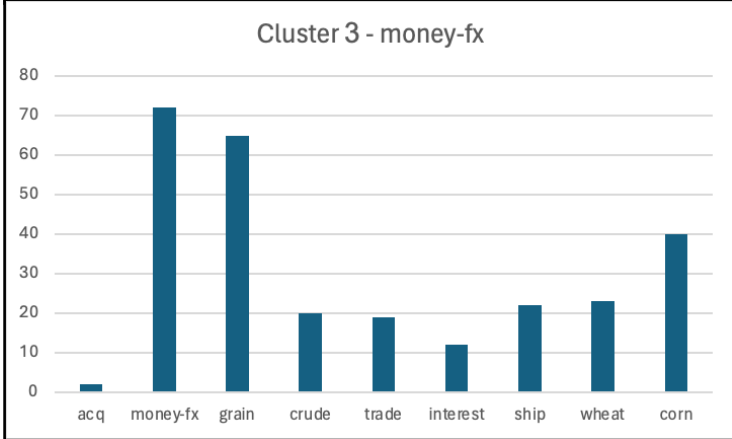
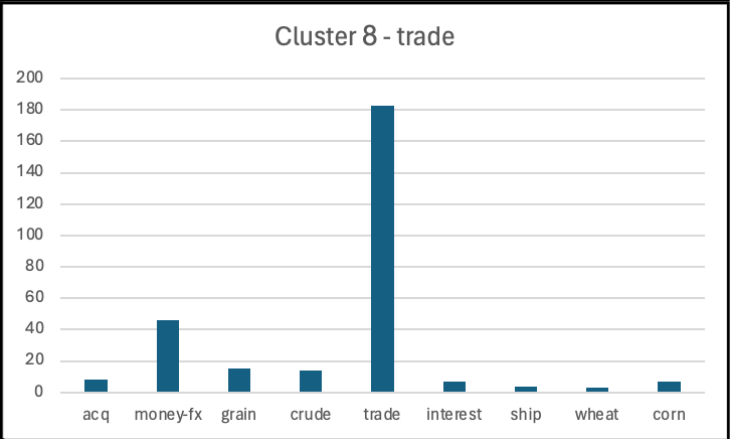
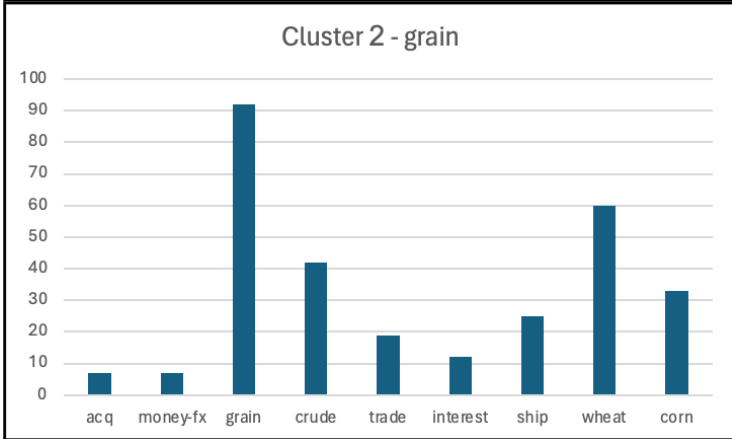
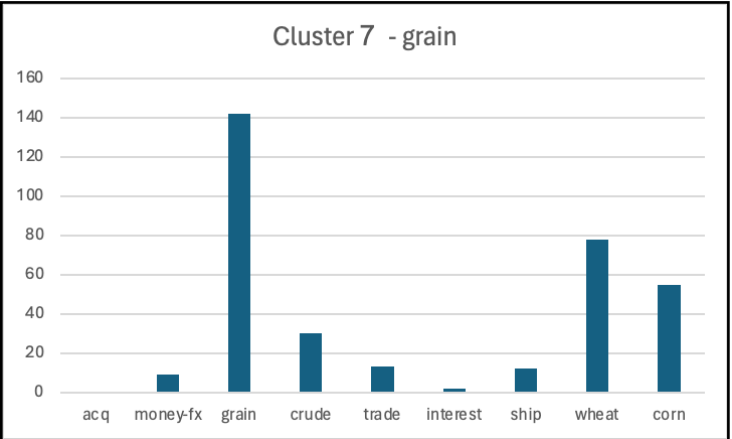
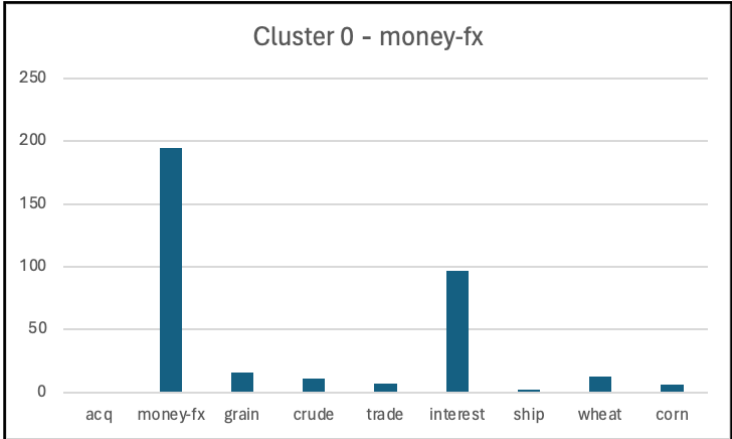
Question 3:

cluster number\topic	acq	money-fx	grain	crude	trade	interest	ship	wheat	corn	cluster size
6	341	6	1	38	6	2	44	1	0	439
5	8	82	13	31	76	127	4	5	5	351
0	0	195	16	11	7	97	2	13	6	347
7	0	9	142	30	13	2	12	78	55	341
2	7	7	92	42	19	12	25	60	33	297
8	8	46	15	14	183	7	4	3	7	287
3	2	72	65	20	19	12	22	23	40	275
1	13	22	35	34	6	8	64	11	11	204
4	21	12	9	129	8	8	9	3	2	201

Question 4:

The 9 histograms are (sorted from largest cluster to smallest):





Question 5:

The accuracy of the classification in my model is 0.633.

Question 6:

The constants used when running the algorithm are:

- K (underflow treatment constant) - 10 (this value was recommended in the instructions).
- Vocabulary size after filtering - 6800.
- Smoothing Lambda - 0.061 (I chose that value with many automated trial and error runs).
- Alpha Epsilon (for preventing the alpha values from reaching zero) - 0.001

* The code is attached inside the zip, I put a lot of effort in it :)